feature representations for a wide range of images. The network has an image input size of 299-by-299. The model extracts general features from input images in the first part and classifies them based on those features in the second part.

In addition, TensorFlow can also be used with containerisation tools such as Docker. For instance, it can be used to deploy a sentiment analysis model that uses character level ConvNet networks for text classification. TensorRec is another cool recommendation engine framework in TensorFlow. It has an easy API for training and prediction, and resembles common machine learning tools in Python.

Tesseract OCR and its execution process

Tesseract is a neural network based OCR engine. It supports more than 100 languages to be recognised from images like png, jpg and tiff, and generates output in various formats like HTML, pdf and plain text, to name a few. It is a library that can be used with other applications, and can also be used as a standalone command-line tool with no facility as built-in GUI.

There are many third-party tools available to be used as Tesseract integrated GUI applications. Tesseract OCR uses third party Leptonica library under BSD-2 clause license to support image recognition in zlib, png and tiff formats.

Tesseract OCR takes an input image from a scanned source and gives it for pre-processing, where it is cleansed (from distortion) to make it as clear as possible using possible pixelation and anti-pixelation methods, resizing for better readability and interpretation by the algorithm, and rotation (if needed) for proper character interpretation.

Then the OCR engine uses suitable algorithms including Leptonica library to start processing the image by splitting it into logical lines of text. Each line of text is then split into words using space as a breaker, and then words are split into characters using logical shapes. This is then used to interpret the character, word and sentence. The final interpreted group of characters is given to the post processor, which will form the words and sentence as per split size of the source image.

During the entire processing activity, Tesseract OCR engine uses its trained data set which keeps on building its neural network whenever reprocessing and processing corrections are carried out. More and more usage and corrections help to build better accuracy.

Tesseract 4.0, the latest version, has improved text line recognition using line separator interpretation from line to line gaps. In general, Tesseract uses adaptive binarization for character recognition from binary format of images; for images with single character, it uses convolutional neural networks (CNN).

Tesseract was developed using C++ and if you want to integrate it with Python applications, you can use Pytesseract (Python Tesseract) which is a Python based wrapper for the Tesseract OCR library. Tesseract GitHub has an in-built facility for language detection when it gets a scanned image to load the appropriate language training set for processing. The training set for different languages is available in GitHub in the *Data-Files* folder as a *traindata* file and placed in *$TESSDATA_PREFIX* folder in the Tesseract installed location.

This finishes with our overview of two powerful AI tools — TensorFlow and Tesseract OCR. END

References

- https://www.educba.com/tensorflow-architecture/
- https://www.tensorflow.org/tfx/serving/architecture
- https://techvidvan.com/tutorials/tensorflow-applications/
- https://github.com/mozilla/DeepSpeech
- https://backlinko.com/google-rankbrain-seo
- https://www.kaggle.com/code/imsparsh/large-categories-image-classifier-inception-v3/notebook
- https://tesseract-ocr.github.io/
- https://github.com/tesseract-ocr/tesseract

By: Dr Anand Nayyar and Dr Magesh Kasthuri

Dr Anand Nayyar is a PhD in wireless sensor networks and swarms intelligence. He works at Duy Tan University, Vietnam, and loves to explore open source technologies, IoT, cloud computing, deep learning and cyber security.

Dr Magesh Kasthuri is a senior distinguished member of the technical staff and principal consultant at Wipro Ltd. This article expresses his views and not that of Wipro.